Chapter 2 - The Real Number System

Note 3

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2.3 : THE COMPLETENESS AXIOM

2.3.1 : The Real Line

Recall that a convenient and familiar geometric interpretation of the real number system is the real line. In this interpretation, the absolute value $|a|$ of an element $a \in \mathbb{R}$ is regarded as the distance from a to the origin 0. More generally, the **distance** between elements a and b in $\mathbb{R}$ is $|a - b|$ (See Figure 2.3.1).

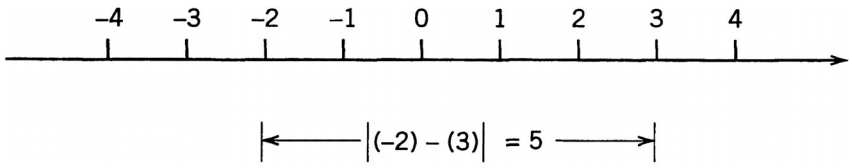


Figure 2.3.1

Later we will need precise language to discuss the notion of one real number being "close to" another. If a is a given real number, then saying that a real number x is "close to" a should mean that the distance $|x - a|$ between them is "small". A context in which this idea can be discussed is provided by the terminology of *neighborhoods*, which we now define.

Definition 1 : Let $a \in \mathbb{R}$ and $\varepsilon > 0$. Then the ε-**neighborhood** of a is the set $V_\varepsilon(a) := \{x \in \mathbb{R} \ : \ |x - a| < \varepsilon\}$.

For $a \in \mathbb{R}$, the statement that x belongs to $V_\epsilon(a)$ is equivalent to either of the statements:

$$-\varepsilon < x - a < \varepsilon \iff a - \varepsilon < x < a + \varepsilon$$

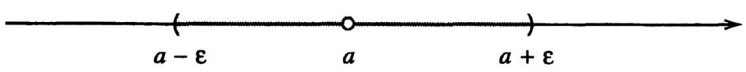


Figure 2.3.2

Lemma : If $a \in \mathbb{R}$ is such that $0 \leq a < \epsilon$ for every $\epsilon > 0$, then $a = 0$.

Proof.

- Suppose to the contrary that $a > 0$.
- Then if we take $\varepsilon_0 := \frac{1}{2}a$, we have $0 < \varepsilon_0 < a$.
- Therefore, it is false that $a < \varepsilon$ for every $\epsilon > 0$ and we conclude that $a = 0$.

□

Remark : It is an exercise to show that if $a \in \mathbb{R}$ is such that $0 \leq a \leq \epsilon$ for every $\varepsilon > 0$, then $a = 0$.

Theorem 2.4 : Let $a \in \mathbb{R}$. If x belongs to the neighborhood $V_\varepsilon(a)$ for every $\varepsilon > 0$, then $x = a$.

Proof. If a particular x satisfies $|x - a| < \varepsilon$ for every $\varepsilon > 0$, then it follows from the Lemma above that $|x - a| = 0$, and hence $x = a$. □

Example 1 : If $|x - a| < \varepsilon$ and $|y - b| < \varepsilon$, then the **Triangle Inequality** implies that

$$\begin{aligned} |(x + y) - (a + b)| &= |(x - a) + (y - b)| \\ &\leq |x - a| + |y - b| < 2\varepsilon \ . \end{aligned}$$

Thus if x, y belong to the ε-neighborhoods of a, b, respectively, then $x+y$ belongs to the 2ε-neighborhood of $a + b$ (but not necessarily to the ε-neighborhood of $a + b$).

2.3.2 : The Completeness Property of $\mathbb{R}$

Thus far, we have discussed the algebraic properties and the order properties of $\mathbb{R}$. In this section, we shall present one more property of $\mathbb{R}$ that is often called the "**Completeness Property**".

The system $\mathbb{Q}$ of rational numbers also has the algebraic and order properties described in the preceding section, but we have seen that $\sqrt{2}$ *cannot* be represented as a rational number; therefore $\sqrt{2}$ does not belong to $\mathbb{Q}$. This observation shows the necessity of an additional property to characterize the real number system. This additional property, the Completeness (or the Supremum) Property, is an essential property of $\mathbb{R}$, and we will say that $\mathbb{R}$ is a *complete ordered field*. It is this special property that permits us to define and develop the various limiting procedures that will be discussed in the chapters that follow.

There are several different ways to describe the Completeness Property. We choose to give what is probably the most efficient approach by assuming that each non-empty bounded subset of $\mathbb{R}$ has a supremum.

Definition 2 : Let $E \subset \mathbb{R}$ be non-empty.

1. The set E is said to be **bounded above** if and only if there is an $M \in \mathbb{R}$ such that $a \leq M$ for all $a \in E$, in which case M is called an *upper bound* of E.
2. A number s is called a **supremum** of the set E if and only if s is an upper bound of E and $s \leq M$ for all upper bounds M of E. (In this case we shall say that E has a *finite supremum* s and write $s = \sup E$.)

By **Definition 2 (ii)**, a supremum of a set E (when it exists) is the smallest (or **least**) upper bound of E. By definition, then, in order to prove that $s = \sup E$ for some set $E \subset \mathbb{R}$, we must show two things: s is an upper bound, and s is the smallest upper bound. Here is a typical example.

Example 2 : If $E = [0, 1]$, prove that $\sup E = 1$.

Proof.

- By the definition of interval, 1 is an upper bound of E.
- Let M be any upper bound of E; that is, $M \geq x$ for all $x \in E$.
- Since $1 \in E$, it follows that $M \geq 1$. Thus 1 is the smallest upper bound of E.

□

The following two remarks answer the question: **How many upper bounds and suprema can a given set have?**

Remark 1 : If a set has one upper bound, it has infinitely many upper bounds.

Proof. If M_0 is an upper bound of a set E, then so is M for any $M > M_0$. □

Remark 2 : If a set has a supremum, then it has only one supremum.

Proof. Let s_1 and s_2 be suprema of the set E. Then both s_1 and s_2 are upper bounds of E, whence by **Definition 2 (ii)**, $s_1 \leq s_2$ and $s_2 \leq s_1$. We conclude by the **Trichotomy Property** that $s_1 = s_2$.

□

Note : This proof illustrates a general principle. When asked to prove $a = b$, it is often easier to verify that $a \leq b$ and $b \leq a$ separately.

The next result, a fundamental property of suprema, shows that the supremum of a set E can be approximated by a point in E (see Figure 2.3.3 for an illustration).

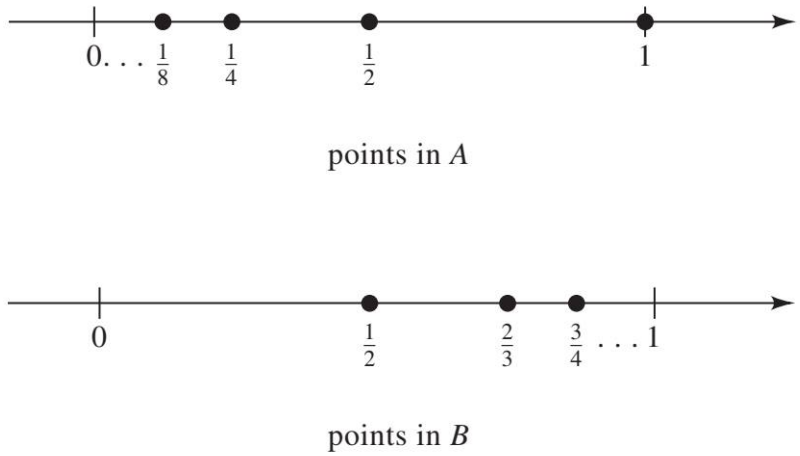


Figure 2.3.3

Theorem 2.5 : **Approximation Property for Suprema**
If E has a finite supremum and $\varepsilon > 0$ is any positive number, then there is a point $a \in E$ such that

$$\sup E - \varepsilon < a \leq \sup E$$

Proof of Theorem 2.5 :

- Suppose that the theorem is false. Then there is an $\varepsilon_0 > 0$ such that no element of E lies between $s_0 := \sup E - \varepsilon_0$ and $\sup E$.
- Since $\sup E$ is an upper bound for E, it follows that $a \leq s_0$ for all $a \in E$; that is, s_0 is an upper bound of E.
- Since $\sup E$ is the **least** of those upper bounds of E, from **Definition 2 (ii)**, we have

$$\sup E \leq s_0 = \sup E - \varepsilon_0 \ .$$

- Adding $\varepsilon_0 - \sup E$ to both sides of this inequality, we conclude that $\varepsilon_0 \leq 0$, which is a contradiction.

□

Theorem 2.6 : If $E \subset \mathbb{Z}$ has a supremum, then $\sup E \in E$. In particular, if the supremum of a set, which contains only integers, exists, that supremum must be an integer.

Proof of Theorem 2.6 *:*

- Suppose that $s := \sup E$ and apply the Approximation Property to choose an $x_0 \in E$ such that $s - 1 < x_0 \leq s$.
- If $s = x_0$, then $s \in E$, as promised.
- Otherwise, $s-1 < x_0 < s$ and we can apply the Approximation Property again to choose $x_1 \in E$ such that $x_0 < x_1 < s$.
- Subtract x_0 from this last inequality to obtain $0 < x_1 - x_0 < s - x_0$.
- Since $-x_0 < 1 - s$, it follows that $0 < x_1 - x_0 < s + (1 - s) = 1$.
- Thus $x_1 - x_0 \in \mathbb{Z} \cap (0, 1)$, a contradiction by **Remark 1.1 (iii)** (Ref. page 4 of Note 2).
- We conclude that $s \in E$.

□

The last assumption that we make about $\mathbb{R}$ is the Existence of Suprema.

Completeness Axiom

If E is a non-empty subset of $\mathbb{R}$ that is bounded above, then E has a finite supremum.

We shall use the Completeness Axiom many times. Our first two applications deal with distribution of integers (**Theorem 2.7**) and rationals (**Theorem 2.8**) among real numbers.

Theorem 2.7 : Archimedean Principle

Given real numbers a and b, with $a > 0$, there is an integer $n \in \mathbb{Z}^+$ such that $b < na$.

Proof of Theorem 2.7.

- If $b < a$, set $n = 1$.
- If $a \leq b$, consider the set $E = \{k \in \mathbb{Z}^+ \ : \ ka \leq b\}$. E is nonempty since $1 \in E$.
- Let $k \in E$ (i.e., $ka \leq b$). Since $a > 0$, it follows from the **First Multiplicative Property** that $k \leq b/a$. This proves that E is bounded above by b/a.

- Thus, by the **Completeness Axiom** and **Theorem 2.6**, E has a finite supremum s that belongs to E, in particular, $s \in \mathbb{Z}^+$.
- Set $n = s + 1$. Then $n \in \mathbb{Z}^+$ and (since n is larger than s), n cannot belong to E. Thus $na > b$. □

Note : In **Example 2** and **Theorem 2.6** that the supremum of E belonged to E. The following result shows that this is not always the case.

Example 3 : Let $A = \left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\right\}$ and $B = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\right\}$. Prove that $\sup A = \sup B = 1$.

Proof.

- It is clear that 1 is an upper bound of both sets. We need to show that 1 is the smallest upper bound of both sets.
- For A, this is trivial. Indeed, if M is any upper bound of A, then $M \geq 1$ (since $1 \in A$).
- On the other hand, if M is an upper bound for B, but $M < 1$, then $1 - M > 0$. In particular, $1/1 - M \in \mathbb{R}$.
- Choose, by the **Archimedean Principle**, an $n \in \mathbb{N}$ such that $n > 1/(1 - M)$. It follows that $x_0 \coloneqq 1 - 1/n > M$.
- Since $x_0 \in B$, this **contradicts** the assumption that M is an upper bound of B. □

The next proof shows how the **Archimedean Principle** is used to establish scale.

Theorem 2.8 : **Density of Rationals**
If $a, b \in \mathbb{R}$ satisfy $a < b$, then there is a $q \in \mathbb{Q}$ such that $a < q < b$.
(i.e., between any two real numbers, there exists a rational number).

Proof.

- Suppose first that $a > 0$. Since $b - a > 0$, use the **Archimedean Principle** to choose an $n \in \mathbb{N}$ that satisfies
$$n > \max\left\{\frac{1}{a}, \frac{1}{b-a}\right\},$$
and observe that both $\frac{1}{n} < a$ and $\frac{1}{n} < b - a$.

- Consider the set $E = \left\{ k \in \mathbb{Z}^+ \ : \ \dfrac{k}{n} \leq a \right\}$. Since $1 \in E$, E is nonempty.
- Since $n > 0$, E is bounded above by na. Hence, by **Theorem 2.6**, $k_0 \coloneqq \sup(E)$ exists and belongs to E, in particular, to $\mathbb{Z}^+$.
- Set $m = k_0 + 1$ and $q = \dfrac{m}{n}$. Since k_0 is the supremum of E, $m \notin E$. Thus $q > a$.
- On the other hand, since $k_0 \in E$, it follows from the choice of n that

$$b = a + (b - a) \geq \frac{k_0}{n} + (b - a) > \frac{k_0}{n} + \frac{1}{n} = \frac{m}{n} = q \ .$$

- Now suppose that $a \leq 0$. Choose, by the **Archimedean Principle**, an integer $k \in \mathbb{Z}^+$ such that $k > -a$.
- Then $0 < k + a < k + b$, and by the case already proved, there is an $r \in \mathbb{Q}$ such that $k + a < r < k + b$.
- Therefore, $q \coloneqq r - k$ belongs to $\mathbb{Q}$ and satisfies the inequality $a < q < b$.

□

For some applications, we also need the following concepts.

Definition 3 : Let $E \subset \mathbb{R}$ be non-empty.

1. The set E is said to be **bounded below** if and only if there is an $m \in \mathbb{R}$ such that $a \geq m$ for all $a \in E$, which case m is called a *lower bound* of the set E.
2. A number t is called an **infimum** of the set E if and only if t is a lower bound of E and $t \geq m$ for all lower bounds m of E. In this case we shall say that E has an *infimum* t and write $t = \inf E$.
3. A set E is said to be *bounded* if and only if it is bounded both above and below.
4. A set is said to be unbounded if it is not bounded.

Note :

- When a set E contains its supremum (respectively, its infimum) we shall frequently write $\max E$ for $\sup E$ (respectively, $\min E$ for $\inf E$).

- The **supremum** is also called **the least upper bound** and the **infimum** is also called **the greatest lower bound**.

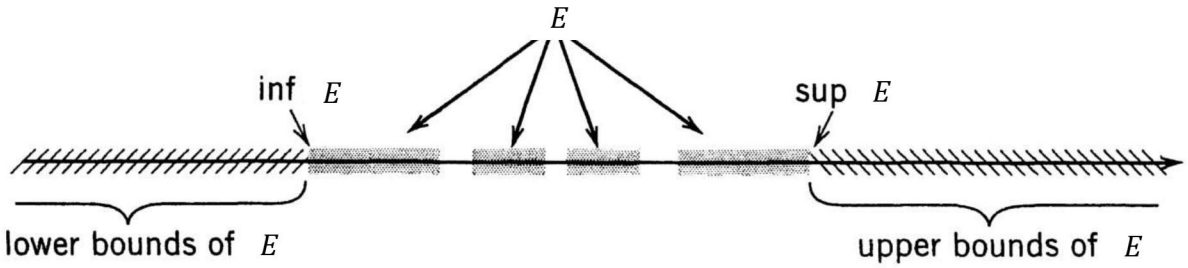


Figure 2.3.4 : $\inf E$ **and** $\sup E$

To relate suprema to infima, we define the **reflection** of a set $E \subseteq \mathbb{R}$ by

$$-E \coloneqq \{x \ : \ x = -a \ \text{ for some } \ a \in E\} \ .$$

For example, $-(1,2] = [-2,-1)$.

Example 4 :

1. The set $S = \{2,4,6,8\}$ is bounded above by $8, 9, 8\frac{1}{2}, \pi^2$, and any other real number greater than or equal to 8. Since $8 \in S$, we have $\max S = 8$. Similarly, S has many lower bounds, including 2, which is the largest of the lower bounds and the minimum of S (See Figure 2.3.5). It is easy to see that any finite set is bounded and always has a maximum and a minimum.

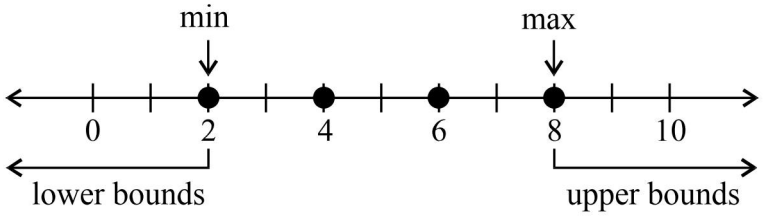


Figure 2.3.5 : $S = \{2,4,6,8\}$

2. The interval $[0,\infty)$ is not bounded above. It is bounded below by any non-positive number, and of these lower bounds, 0 is the largest. Since $0 \in [0,\infty)$, 0 is the minimum of $[0,\infty)$ (See Figure 2.3.6).

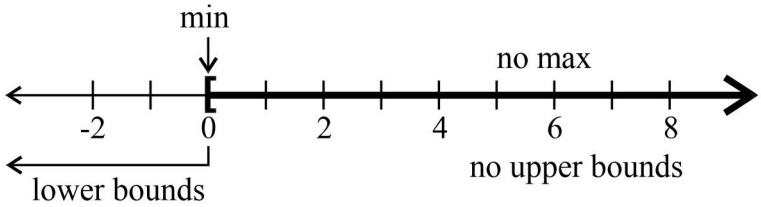


Figure 2.3.6 : $S = [0,\infty)$

3. The interval $(0, 4]$ has a maximum of 4, and this is the smallest of the upper bounds. It is bounded below by any nonpositive number, and of these lower bounds, 0 is the largest. Since $0 \notin (0, 4]$, the set has no minimum (See Figure 2.3.7).

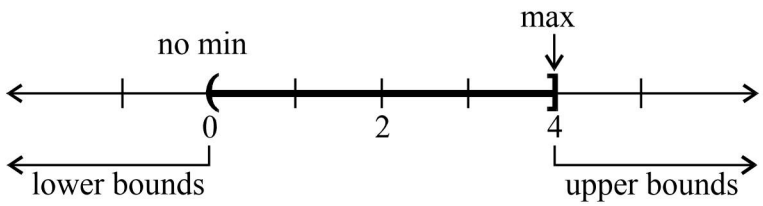


Figure 2.3.7 : $S = (0, 4]$

4. The empty set $\emptyset$ is bounded above by any $m \in \mathbb{R}$. (The condition $m \geq s$ for all $s \in \emptyset$ is equivalent to the implication "if $s \in \emptyset$, then $m \geq s$". This implication is true since the antecedent is false.) Likewise, $\emptyset$ is bounded below by any real m.

5. Any subset of a bounded set will also be bounded. Indeed, if $T \subseteq S$ and m is an upper bound of S, then m will also be an upper bound of T. The same holds for lower bounds.

The following result shows that the supremum of a set is the same as the negative of its reflection's infimum. This can be used to prove an Approximation Property and a Completeness Property for Infima (see **Exercise 3**).

Theorem 2.9 : **Reflection Principle**

Let $E \subseteq \mathbb{R}$ be non-empty.

(i). E has a supremum if and only if $-E$ has an infimum, in which case

$$\inf(-E) = -\sup(E)$$

(ii). E has an infimum if and only if $-E$ has a supremum, in which case

$$\sup(-E) = -\inf(E)$$

Proof of Theorem 2.9.

- The proofs of these statements are similar. We prove only the first statement. In your notes, please write a proof for **ii**.

- Suppose that E has a supremum s and set $t = -s$. Since s is an upper bound for E, $s \geq a$ for all $a \in E$, so $-s \leq -a$ for all $a \in E$. Therefore, t is a lower bound of $-E$.

- Suppose that m is any lower bound of $-E$. Then $m \leq -a$ for all $a \in E$, so $-m$ is an upper bound of E.
- Since s is the supremum of E, it follows that $s \leq -m$ (i.e., $t = -s \geq m$). Thus t is the infimum of $-E$ and $\sup E = s = -t = -\inf(-E)$.
- Conversely, suppose that $-E$ has an infimum t. By definition, $t \leq -a$ for all $a \in E$. Thus $-t$ is an upper bound for E. Since E is nonempty, E has a supremum by the **Completeness Axiom**.

□

Theorem 2.9 allows us to obtain information about infima from results about suprema, and vice versa (see the proof of the next theorem).

Theorem 2.10 : **Monotone Property**

Suppose that $A \subseteq B$ are non-empty subsets of $\mathbb{R}$.

(i). If B has a supremum, then $\sup A \leq \sup B$.

(ii). If B has an infimum, then $\inf A \geq \inf B$.

Proof of Theorem 2.10.

(i).
- Since $A \subset B$, any upper bound of B is an upper bound of A.
- Therefore, $\sup B$ is an upper bound of A. It follows from the **Completeness Axiom** that $\sup A$ exists, and from **Definition 2 : (ii)** that $\sup A \leq \sup B$.

(ii). Clearly, $-A \subseteq -B$. Thus by part (i), **Theorem 2.9**, and the Second Multiplicative Property, it follows that

$$\inf A = -\sup(-A) \geq -\sup(-B) = \inf B \ .$$

□

Remark 1 :

- It is convenient to extend the definition of suprema and infima to all subsets of $\mathbb{R}$. To do this we expand the definition of $\mathbb{R}$ as follows.
- The set of Extended Real Numbers is defined to be $\overline{\mathbb{R}} \coloneqq \mathbb{R} \cup \{\pm\infty\}$. Thus x is an extended real number if and only if either $x \in \mathbb{R}$, $x = +\infty$, or $x = -\infty$.

Remark 2 :

- Let $E \subseteq R$ be nonempty. We shall define $\boxed{\sup E = +\infty}$ if E is **unbounded above** and $\boxed{\inf E = -\infty}$ if E is **unbounded below**.
- We define $\boxed{\sup \emptyset = -\infty}$ and $\boxed{\inf \emptyset = +\infty}$, where recall that $\emptyset$ denotes the empty set.
- Notice that the supremum of a subset E of $\mathbb{R}$ (respectively, the infimum of E) is finite if and only if E is nonempty and bounded above (respectively, nonempty and bounded below).
- Moreover, under the convention $-\infty < a$ and $a < \infty$ for all $a \in \mathbb{R}$, the Monotone Property still holds for this extended definition; that is, if A and B are subsets of $\mathbb{R}$ and $A \subseteq B$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$, provided we use the convention that $-\infty < \infty$.

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EXERCISES

1. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.

   (a). If A and B are nonempty bounded subsets of $\mathbb{R}$, then $\sup\,(A \cap B) \leq \sup\,A$.

   (b). Let ε be a positive real number. If A is a nonempty, bounded subset of $\mathbb{R}$ and $B = \{\varepsilon x : x \in A\}$, then $\sup(B) = \varepsilon\,\sup\,(A)$.

   (c). If $A + B := \{a + b : a \in A \text{ and } b \in B\}$, where A and B are nonempty. bounded subsets of $\mathbb{R}$, then $\sup(A + B) = \sup(A) + \sup(B)$.

   (d). If $A - B := \{a - b : \text{a} \in A \text{ and } b \in B\}$, where A and B are nonempty bounded subsets of $\mathbb{R}$, then $\sup(A - B) = \sup(A) - \sup(B)$.

2. Find the infimum and supremum of each of the following sets

   (a). $E = \{x \in \mathbb{R} \ : \ x^2 + 2x = 3\}$

   (b). $E = \{x \in \mathbb{R} \ : \ x^2 - 2x + 3 > x^2 \text{ and } x > 0\}$

   (c). $E = \{p/q \in \mathbb{Q} \ : \ p^2 < 5q^2 \text{ and } p, q > 0\}$

   (d). $E = \{x \in \mathbb{R} \ : \ x = 1 + (-1)^n/n \text{ for } n \in \mathbb{Z}^+\}$

   (e). $E = \{x \in \mathbb{R} \ : \ x = 1/n + (-1)^n \text{ for } n \in \mathbb{Z}^+\}$

   (f). $E = \{2 - (-1)^n/n^2 \ : \ n \in \mathbb{Z}^+\}$

3. Use the **Reflection Principle** and analogous results about suprema to prove the following results:

   (a). **[Approximation Property for Infima]** Prove that if a set $E \subset \mathbb{R}$ has a finite infimum and $\epsilon > 0$ is any positive number, then there is a point $a \in E$ such that $\inf E + \epsilon > a \geq \inf E$.

   (b). **[Completeness Property for Infima]** If $E \subseteq \mathbb{R}$ is nonempty and bounded below, then E has a (finite) infimum.